

Fig. 3: An actual-size, single-side PCB for the hybrid solar charger

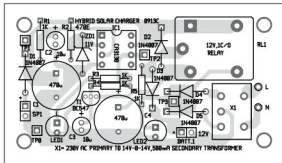


Fig. 4: Component layout for the PCB

high and the same is indicated by glowing green LED1. Transistor T1 then conducts and relay RL1 energizes. Thus the battery gets charging current from the solar panel through the normally-open (N/O) and common contacts of relay RL1.

LED2 indicates charging of the battery. Capacitor C3 is provided for clean switching of transistor T1. Diode D2 protects T1 from back EMF and diode D3 prevents the discharge of battery current into the circuit.

When output from the solar panel drops below 12 volts, output of the comparator turns low and the relay de-energizes. Now the battery gets charging

current from the transformer-based power supply through the normally-

closed (N/C) and common contacts of the relay. This power supply comprises step-down transformer X1, rectifying diodes D4 and D5, and smoothing capacitor C4.

Construction and testing

An actual-size, single-side PCB for the hybrid solar charger is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Use high-gauge (thick) wires to connect the solar panel and the battery to the circuit.

To test the circuit for proper functioning, remove the solar panel from connector SP1 and connect a DC variable voltage source. Set some voltage below 12V and slowly increase it. As the voltage reaches 12V and goes beyond, the logic at test point TP2 changes from low to high. The transformer-based power supply voltage can be checked at test point TP3. ●

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input gets a higher voltage at this time, the output of the comparator turns

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